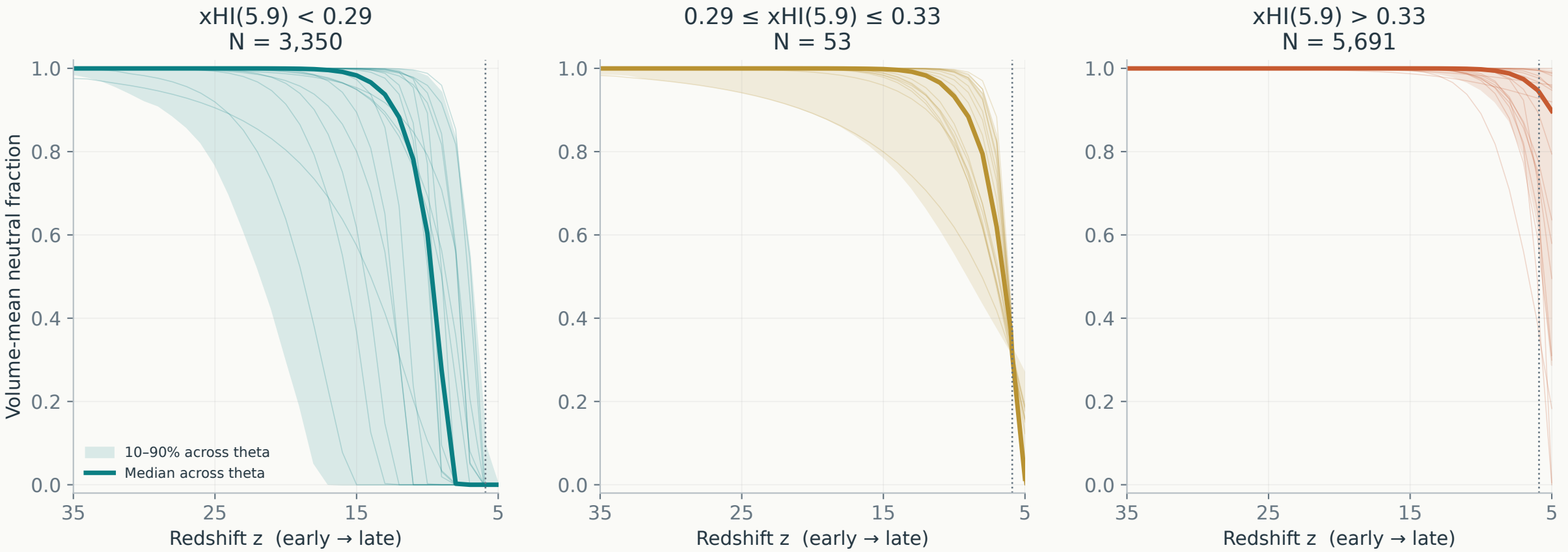
# 01 | Reionization history diversity

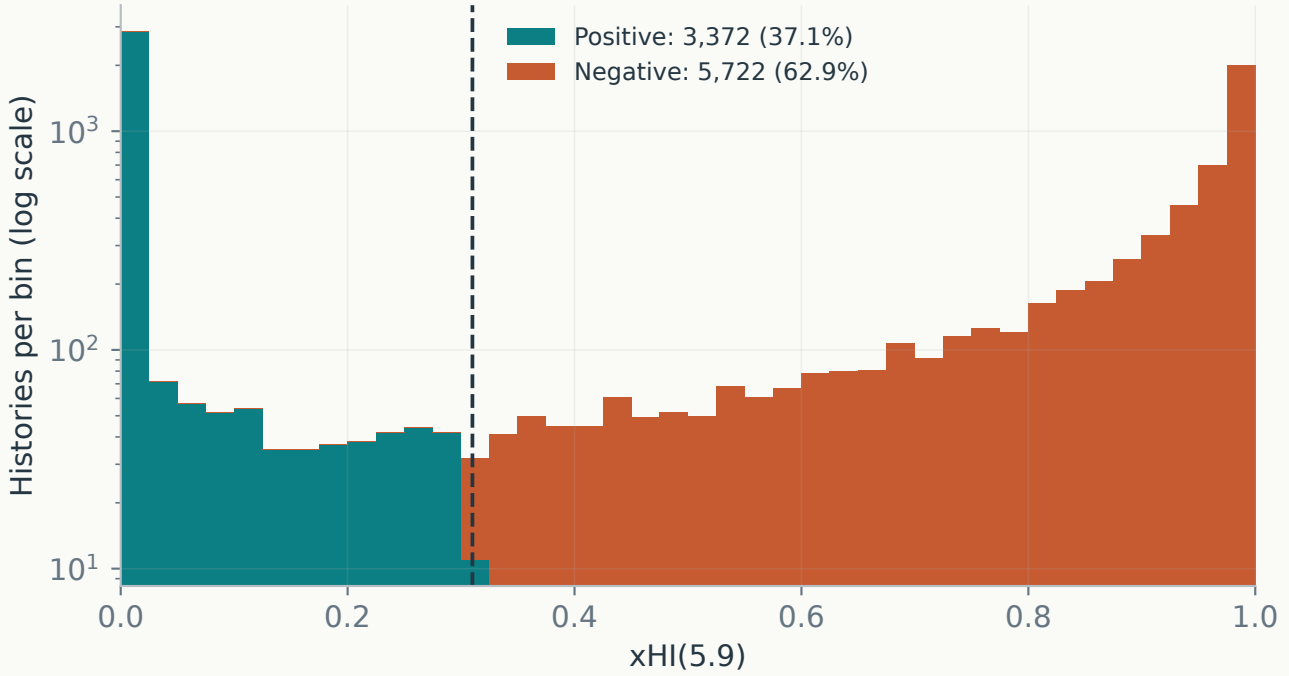
Frozen random-IC Train snapshot: 9,094 histories. Original 32-node arrays; no smoothing.

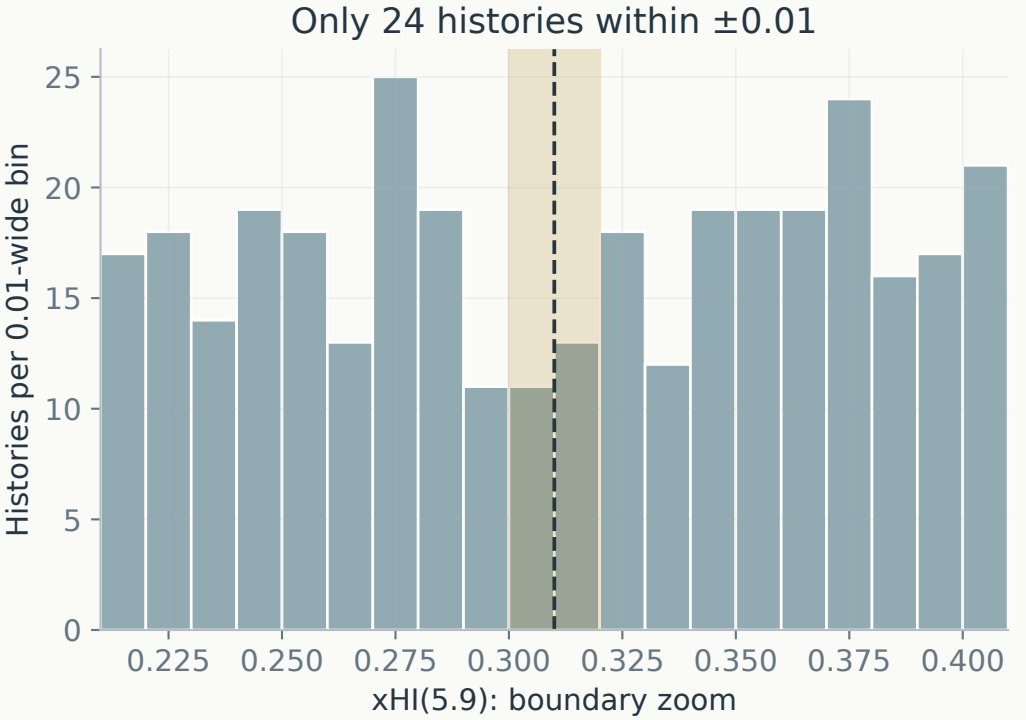



Thin lines: 18 reproducibly sampled exact histories/panel. Bands describe different theta, not IC scatter at fixed theta.

# 02 | Classifier balance and the boundary

Cut = xHI(5.9) < 0.31, from the frozen original likelihood. Negative histories remain in the dataset.



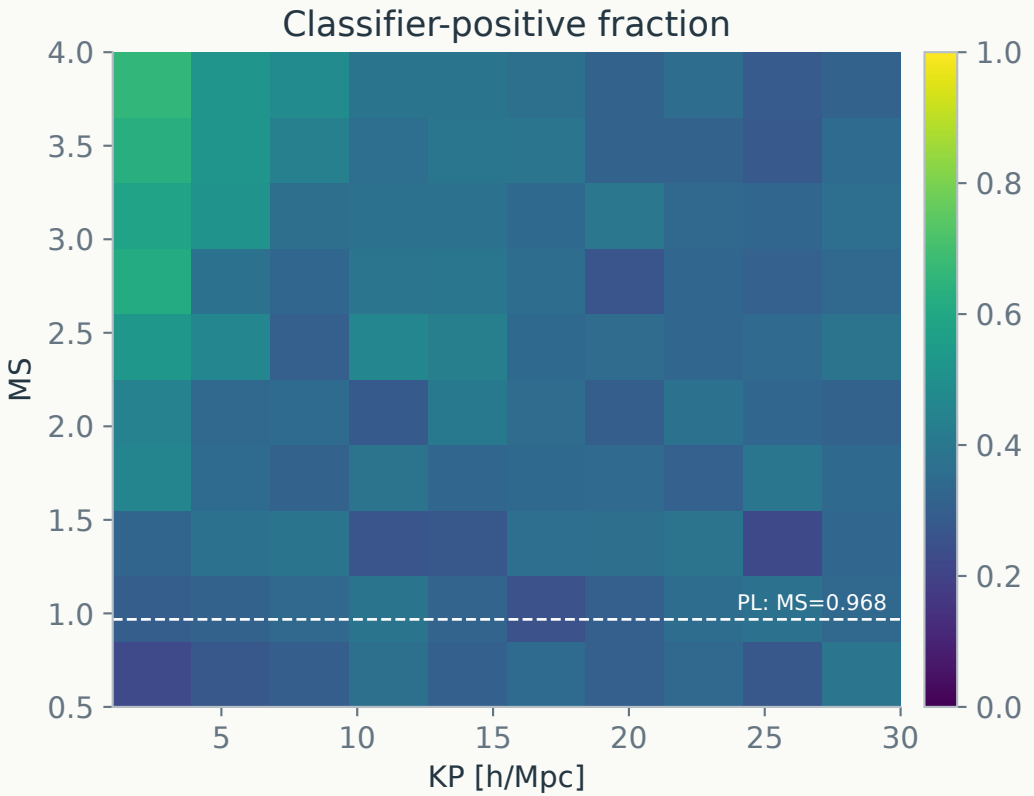



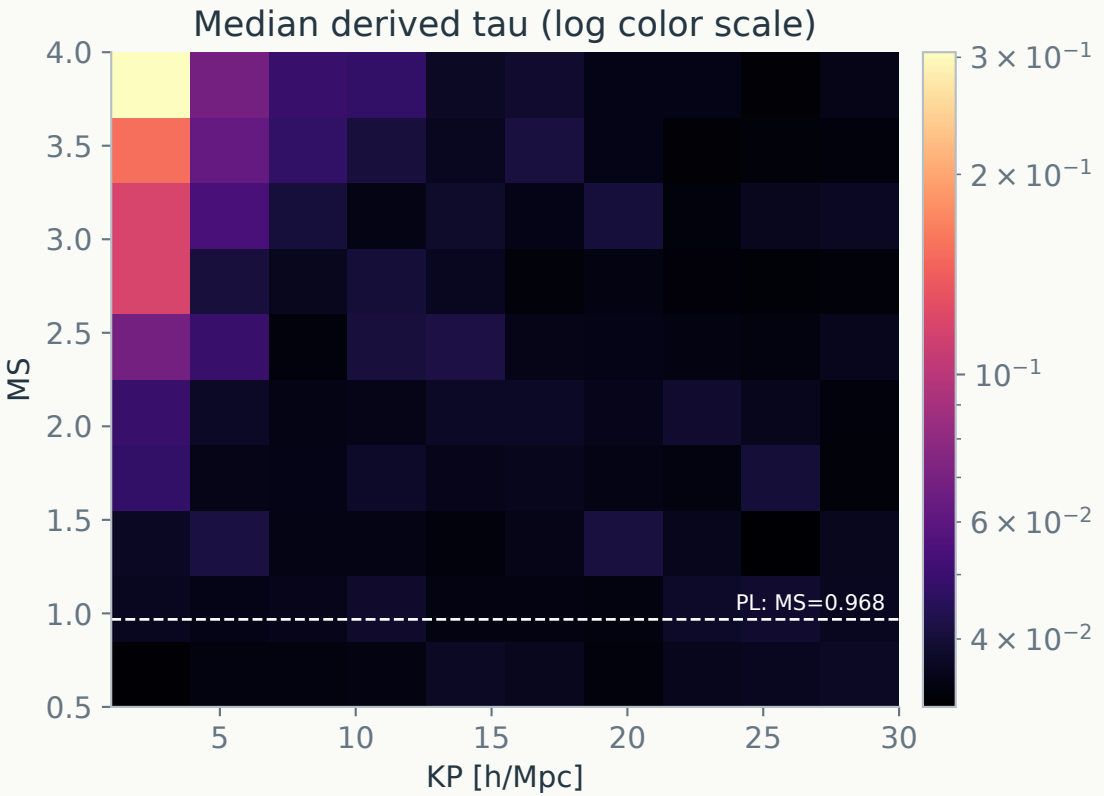

1,945 / 3,372 positives also have logL_tau ≥ −12.5. This excludes LF and is not posterior acceptance.

Counts are from the mixed training design, not the scientific prior. Cut remains unchanged; no classifier hard gate enabled.

# 03 | KP / MS structure in the broad sample

Broad stratum only: 7,277 histories; 10 × 10 bins, 62–84 histories/bin. Other astrophysical parameters vary.

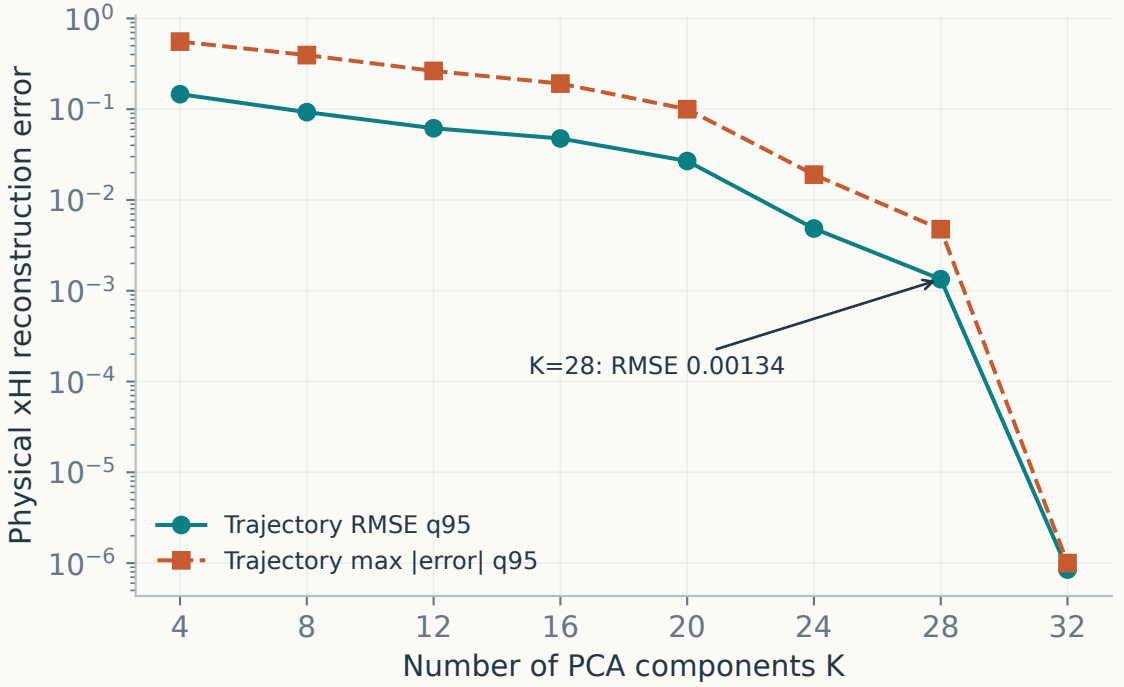


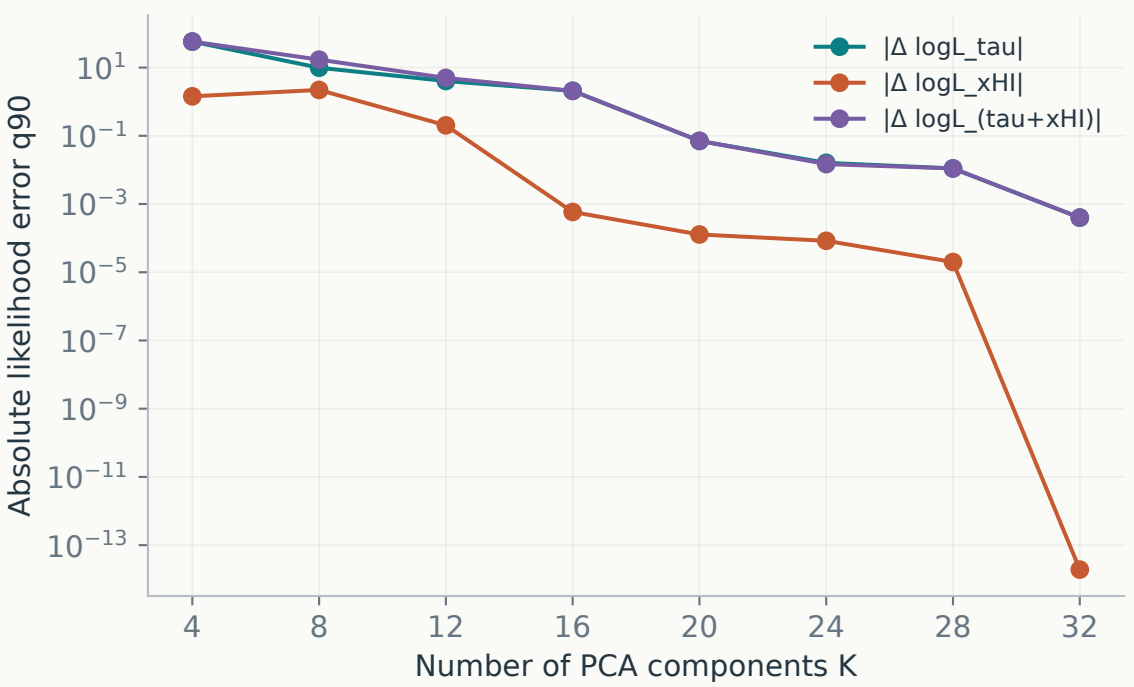




Descriptive associations in the broad design. These maps are neither BT-only causal effects nor KP/MS posterior constraints.

# 04 | How much PCA compression is plausible?

Logit PCA fitted to 3,372 Train-positive histories. Exact coefficients; these are in-sample reconstruction errors, not NN errors.




K=28 reduces 32 coefficients to 28 (12.5%). Independent Validation is still required before choosing a decoder.

History errors: all 3,372 positives. Likelihood errors: the same fixed 256-point subset, using original postprocessing; epsilon=1e−6.